

Emmanuel Acheampong Oppong

3182454129 | Monroe, Louisiana | emmanuelopponga07@gmail.com

therealemmanueloppong.github.io | linkedin.com/in/emmanuel-oppong-acheampong | github.com/therealemmanueloppong

EDUCATION

Grambling State University

Bachelor of Science in Computer Science

Relevant Coursework: Object-Oriented Programming, Software Engineering, Data Structures & Algorithms, Operating Systems, Computer Networks, Cloud Computing

Awards: 3.95/4.0 GPA, Dean's List (2024, 2025), **Presidential Scholarship**

Expected May 2028

Grambling, Louisiana

TECHNICAL SKILLS

Languages: Python, Java, Go, C, C++, TypeScript/JavaScript, SQL

Technologies: Kafka, Redis, PostgreSQL, Docker, Kubernetes, Linux, AWS, React, Next.js, Node.js, Spring Boot, FastAPI, Flask, TensorFlow, PyTorch, LangChain, LlamaIndex, HuggingFace, Scikit-learn, Pandas, Firebase, MongoDB

Tools: Git/GitHub, Bash, CI/CD, CircleCI, Bazel, Pytest

Concepts: Data Structures & Algorithms, Distributed Systems, Microservices, REST APIs, Event-Driven Systems

EXPERIENCE

AI4ALL

Jun 2026 – Aug 2026

Ignite Fellow — Remote

- Built a **financial NLP system** predicting next-day stock price direction from **1.4M headlines and 20.9M stock records**.
- Built an **end-to-end pipeline** covering data cleaning, temporal alignment, bigram/trigram NLP feature extraction, and **FINBert sentiment classification** on 50K headlines using **HuggingFace Transformers and Scikit-learn**.
- Trained a **Random Forest classifier** with **54.1% mean 5-fold cross-validation accuracy**; evaluated Logistic Regression and FINBert baselines to compare LLM-derived signals against traditional ML.
- Presented findings and **model trade-offs** in weekly fellow cohort reviews, incorporating mentor feedback to refine **feature engineering** choices and strengthen the final report.

Bismuth Technologies (Java Platform)

May 2024 – Aug 2025

Software Engineer Intern — Accra, Ghana

- Built an **LLM-powered monitoring pipeline** using **JFR telemetry** to detect subtle regressions across critical Java services.
- Reduced LLM context and inference cost by **>95%**, compressing ~43K JFR data tokens to ~2K through visualization.
- Automated **root-cause analysis** by using internal MCP tools to correlate logs, reducing hours of manual investigation.
- Integrated Slack alerts**, delivering service reports with embedded Grafana links, helping on-calls triage issues faster.

University of Ghana, Legon

Nov 2022 – Aug 2023

Software Engineer (R&D) — Accra, Ghana

- Developed **RAG pipelines** integrating retrieval models with LLMs, achieving ~80% accuracy on complex search queries.
- Co-developed a large-scale **graph visualization tool** enabling 10K+ node exploration for social-network analysis.
- Reduced Docker image size by ~35%** using multi-stage builds, improving deployment speed and lowering AWS costs.
- Automated Docker container restarts** using bash scripts triggered by cron, preventing hours of monthly downtime.

Kehilah Global (API Team)

May 2022 – Aug 2022

Software Engineer Intern — Remote

- Built a backend service using **Flask** to significantly accelerate on-call engineers' response to deployment issues.
- Implemented a singleton database class** to manage SQL connections, significantly reducing connection overhead.
- Achieved ~98% test coverage** with Pytest unit and integration tests, ensuring API reliability and stability.
- Delivered regular sprint updates, enabling faster iteration through rapid feedback loops and timely ticket resolution.

PROJECTS

Stock Price Prediction (AI4ALL)

github.com/jiaying-l/AI4All_Project

- Built an **end-to-end financial NLP pipeline** over **322K headlines and 20M stock records** using **FINBert sentiment analysis**.
- Designed a novel **Competitive Pressure Index** capturing negative news surge relative to each stock's baseline coverage; trained a **Random Forest classifier** achieving **54% 5-fold cross-validated accuracy** and a Logistic Regression baseline with full confusion matrix evaluation.

BuukEasy — Next.js, PostgreSQL, Tailwind CSS, Vercel

- Built a **booking platform** with mobile-first UI and scheduling workflows for solo beauty and wellness professionals.
- Launched MVP** for client booking and scheduling, serving 5 clients and iterating on features based on user feedback.

CITIES Dashboard — React, Material UI, Google Charts, GitHub Actions

- Co-developed an **interactive dashboard** providing 20K+ students across Accra schools with air quality insights.
- Automated deployment workflows** using GitHub Actions, streamlining CI/CD and reducing manual deployment effort.

LEADERSHIP & PROFESSIONAL DEVELOPMENT

ColorStack Academic Chair 2026/2027 (GSU) • AI4ALL Ignite Fellow 2026 • JPMorgan Data for Good Hackathon (April 2026) • Open-Source: Apache Kafka, pgjdbc, kubernetes-client/java, facebook/react (20+ PRs) • NSBE Treasurer • Lantern Network Ambassador